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ORIGINAL ARTICLE

Optimal three-part tariff pricing with Spence-Mirrlees reservation prices

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Abstract
We study the pricing of three-part tariffs (3PTs), where service providers charge a fixed fee with an allowance of free units, and a per-unit fee for additional units above the allowance. This is a pricing strategy that has been widely used in a variety of industries, including telecommunications and internet services. Many papers study the properties of the optimal solution of 3PTs. However, the existing solution approaches proposed in prior papers can only be used to solve 3PT problems under certain restricted conditions. Solving 3PT problems with multiple consumer types has been considered as "intractable" in the literature. In this paper, we provide an optimal solution for 3PT problem with multiple consumer types. We first provide a solution approach to solve 3PT problem with discrete units and multiple discrete consumer types whose willingness-to-pays for different units follow Spence-Mirrlees single-crossing property. The cost of our solution approach is low-rank polynomial to the problem size. We then extend the discrete solution to solve 3PT problems when the consumer types and unit amounts are not required to be discrete. The solution values for the continuous problems are within arbitrary prespecified error tolerances.

Keywords Pricing · Three-part tariff · Non-linear pricing · Price discrimination

1 Introduction

Three-part tariffs have wide applications in business and many consumer markets, where consumers desire several units of a service or product. Examples of such markets

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are wireless and mobile industry, membership discount services, cloud-based data services, credit cards, and car leasing contracts among others. In a three-part tariff (3PT) the vendor charges a fixed access price F (access price) for Q free units (usage allowance), and a per unit charge s above the number of free units. This means that the consumer pays the access price F at the beginning and will get Q units for free. If the consumer purchases more than Q units, she will pay s for each additional unit. Car leasing contracts commonly allows 12,000 miles per year (Q) and then charge around 15 cents (s) per mile over the amount allowed in the contract. Some golf clubs charge a fixed access price F and then another fee s per golf rounds played, after a certain number of rounds played (Q). Software-as-a-Service (SaaS) is a mean of a software delivery that enables data to be accessed from any device. Through SaaS model, software is licensed on a subscription basis. 3PT pricing is being used for a few of the software they offer, including TextMarks, which is a text message platform for businesses. TextMarks charges a base subscription for each of their plans, limiting the number of keywords available on each plan. If a business exceeds this number, they pay for overage.

In the literature, there are quite a few papers that study the optimality of 3PT pricing. Most of those papers focus on revealing the properties of the optimal solution of 3PTs. However, the existing solution approaches proposed in prior papers can only be used to solve 3PT problems under certain restricted conditions, which will be discussed in details in Sect. 1.1. Solving 3PT problems with multiple consumer types has been considered as "intractable" in the literature. For example, in Bhargava and Gangwar (2018), the authors claim that "the 3PT optimization problem with heterogeneous consumers is generally intractable." In the literature, most of the analytical modeling papers model 3PT problems with products on the continuous space, and derive their results based on the first-order or second-order differentiation of the objective functions that maximize the sellers' profits. As discussed in these papers (e.g., Bhargava and Gangwar 2018), 3PT problem is a non-convex optimization problem, and the complexity of the problem increases exponentially with the number of consumer types. Therefore, even though 3PT is well known as a profitable pricing strategy, and has been widely adopted in a variety of different industries, there does not exist any framework to guide firms how to optimally determine the pricing scheme if the firms would like to use 3PT to price their products or services.

In this paper, we provide an optimal solution approach to solve 3PT problems with any number of consumer types. We start to analyze the 3PT problem with discrete products and discrete consumer types. To the best of our knowledge, we are the first paper that studies 3PT problems with such discrete settings. In the discrete space, we discover some critical properties of the optimal solution, which dramatically reduces the size of the feasible set of the optimal solution. With such properties, we develop a solution approach that finds the optimal solution in low-rank polynomial time cost to the problem size. Such a model can be used to guide service providers to optimally determine the 3PT pricing schemes, for discrete type of products or services, such as golf clubs or SaaS.

In addition to solving the 3PT problems with a discrete setting, we revisit the deviations of 3PT models that have been studied in the literature. We first extend our focus to the 3PT model with discrete consumer types and continuous product types. Such a model is

studied in Fibich et al. (2017), where the authors provide an optimal solution with only two consumer types. We remark that our model allows to have any number of consumer types. Based on the solution approach we developed for the discrete case, we find a solution approach for the case with discrete consumer types and continuous product types. For any given error tolerance, our solution approach can generate an approximate solution, such that the gap between the true optimal value and our approximation value is guaranteed to be within the prespecified error tolerance. In addition, the time cost of our approximation solutions are to the low-rank polynomial of the problem size and the error tolerance.

After solving these two types of 3PT problems, we further extend the model to the case when both consumer types and product types are continuous. This is the same model that is studied in Bhargava and Gangwar (2018). Bhargava and Gangwar (2018) claim this model to be "intractable", and focus on providing properties of the optimal solution and deriving conditions on when two-part tariff pricing yields the same optimal profit as the three-part tariff pricing. We extend our discrete solution approach to solve this model with continuous consumer types and product types.

Next, we review the existing studies on 3PT, and discuss what are the existing results in the literature. Among these papers, we discuss three papers that analyze the optimal pricing scheme of 3PT in detail in order to distinguish the unique contributions of our paper.

1.1 Literature review

Several studies analyze the pricing aspect of 3PTs both numerically and empirically. These papers demonstrate substantial evidence for the extensive application of 3PTs in practice. Lambrecht et al. (2007) estimate the demand of a discrete/continuous 3PT model by using a consumer-level Internet usage data set. Even though they do not calculate the optimal solution for 3PTs, they determine that demand uncertainty is a significant factor in solving 3PT problems. According to their empirical analysis, demand uncertainty leads to a decline in consumer surplus and a gain in provider revenue. Iyengar et al. (2007) analyze a 3PT empirically using a wireless service industry data set while capturing for the uncertainly about the quality of the service provider from the viewpoint of consumers and the uncertainty about the consumers' usage. Iyengar et al. (2008) use conjoint analysis to numerically study 3PTs for wireless services. They recognize the two-way dependence between consumption and price. Ascarza et al. (2012) study a market in which consumers switch from 2PT to 3PT using a discrete-continuous model and show consumer over usage due to this change. Chao (2013) analyzes 3PTs under an oligopoly and indicated that product differentiation leads to a more intense competition.

3PTs are being used extensively in heterogeneous markets as opposed to other nonlinear tariff plans. Bhargava and Gangwar (2018) show that 3PTs are more cost effective than per-unit and unlimited-use pricing. In addition, 3PTs are proven to be more powerful than two-part tariff (2PT), in which the vendor charges a fixed access price F and a per unit charge s (Bagh and Bhargava 2013). A few studies analyzed the optimal solution of 2PTs (Essegaier et al. 2002). However, the optimal 3PT has only been cap-

tured under certain limiting circumstances or for prespecified distributions of market demand, and not generally. The pricing aspect of 2PTs and 3PTs, has been studied, both numerically and theoretically, in the literature for several years. There are a few studies in literature, which compare 2PT and 3PT analytically. Among those is Bagh and Bhargava (2013), who show that 3PTs are more efficient than 2PTs under certain market conditions. Baek and Brueckner (2015) and Fibich et al. (2017) show the dominance of a 3PT over 2PT for markets with two segments users.

To the best of our knowledge, there are only three studies, which solve the optimal 3PT pricing problem, all of them under certain conditions. We next discuss each of these three papers in details and highlight the difference between our study to theirs. Grubb (2009) solves the optimal 3PT in the context of cell-phone providers, when consumers overestimate their actual demand. This overestimate means that each consumer has an actual demand and an estimated demand and chose a cell-phone plan based on the estimated demand, which results in a larger profit for the provider. Such an overestimated demand significantly reduces the complexity of the problem and thus leads to a close form solution. Thus, their results are restricted to be applied only to the industries such as telecommunications, where the overestimated demand assumption naturally holds. In our paper, we solve for the situation when the consumers are not assumed to overestimate their demand. Compare to Grubb (2009), since we use the real demands rather than the demands that are required to be overestimated, our results could be applied to many more settings and industries.

Fibich et al. (2017) calculate the optimal 3PT for two-segments consumers markets (heavy and light consumers) with discrete distributions. They innovate an approach to find the optimal solution of 3PT by obtaining the upper bound of the optimal value. In addition, they expand the solution approach to some other conditions, such as consumers incurring psychological costs for non-free units, or having stochastic consumption rates. One important feature of their model is that, they only consider homogeneous consumers or heterogeneous consumers with two types. In our context, they only solve the 3PT problem with one or two consumer types. Therefore, their solution approach cannot be adopted to obtain the optimal pricing scheme when a firm has more than two consumer types. In our model, we solve the problem for multiple consumer types, where the number of consumers can be any number. Since our solution approach can solve problems with large number of consumers efficiently, our solution approach not only works with consumer groups, but also works when the firm wants to treat each consumer individually to further improve profitability.

Bhargava and Gangwar (2018) study the 3PT problem from the theoretical stand point for certain discrete and continuous market demands. According to their analysis, the optimal solutions for the 3PT and 2PT are interchangeable, when the market demand follows an increasing price elasticity. On the other hand, using 3PT is effective when market demand is multimodal or when users have specific tariff biases. We remark that Bhargava and Gangwar (2018) is the only paper that provides solution approaches for 3PT problems with more than two consumer types. They provide a reformulation for the 3PT problem and show that such a reformulation could be used to solve 3PT problems with continuous units and discrete consumer types. When the consumer types are continuous, they demonstrate that they can solve some special cases of 3PT problems with certain specific WTP functions and consumer distribution types.

Please refer to Appendix B.2 in Bhargava and Gangwar (2018) for details. They do not provide guidelines regarding how to solve their reformulation beyond the special WTP functions and the special distribution functions they use. Thus, when both the consumer types and the units are continuous, the optimal solution approach remains unknown for general WTP functions and general distribution functions. In section 4 of our paper, we revisit the model with continuous units and consumer types, the same model that is studied in Bhargava and Gangwar (2018). We relax the restrictions on the specific types of consumer distributions and WTP functions that can be used. We develop a solution approach that can provide an approximate solution to the problem. The gap between the approximation value to the true optimal value is within any prespecified error tolerance.

As a summary, despite all the advances in finding the optimal solution for 3PTs, none of the aforementioned studies were able to find the optimal solution for 3PTs in general. "The 3PT optimization problem with heterogeneous consumers is generally intractable." (see Bhargava and Gangwar 2018, p.1520). Hence, all the previous studies in the literature, which analyzed the 3PT optimization problem had to either make simplifying assumptions on the demand and consumers, or focus on subsets of the problems.

The solution approach proposed in this paper significantly expands the scope of 3TP problems that can be solved efficiently. The critical contribution of this work is to guide firms to optimally setup the pricing scheme when they want to adopt 3PT to price their products or services. Before our work, firms cannot get such guides from prior studies, and thus have to setup a sub-optimal pricing scheme that leads to loss of profit. Furthermore, firms can easily use our model to predict the optimal profit if they adopt 3PT. By comparing to the optimal profits of other nonlinear pricing strategies, such as units selling, 2PT, or other type of nonlinear tariff pricing, firms can determine whether 3PT is the best pricing strategy to adopt.

In the literature, various 3PT models have been considered regarding whether the consumer types are discrete or continuous and whether the products/services are discrete or continuous. Each type of model fits within different industries. For example, golf clubs fit better with the models within discrete products, since they price for the discrete number of rounds played. On the other hand, telecommunication service providers fit better within the models with continuous products, since they charge for the continuous data amounts. Furthermore, each firm may have its own preferences regarding whether to treat each consumer individually, or put consumers into discrete consumer groups, or even treat all the consumers together as one group that follows a certain type of distributions. In this paper, we consider three types of 3PT models and will present these cases separately. The first case is when both the consumer types and the amounts are discrete. To the best of our knowledge, such a model has never been studied in the literature. The second case is when the consumer types are discrete but the amounts are continuous. Fibich et al. (2017) study such model with two consumer types, while we allow the number of consumer types to be a large number. Finally the third case presents a scenario where both the consumer types and the amounts are continuous. Bhargava and Gangwar (2018) analyze such a model (also with Spence-Mirrlees single-crossing property) and provide some properties for the

optimal solution. We present the optimal solution for this problem, which can provide an optimal value within any given error tolerance to the true optimal value.
The rest of the paper is organized as follows: Sect. 2 describes the case in which both the consumer types and the amounts are discrete. Section 3 focuses on the case in which the consumer types are discrete but the amounts are continuous. Section 4 discusses the case in which both the consumer types and the amounts are continuous. Section 5 provides the concluding remarks and possible future extensions.

2 Discrete consumer types and discrete amounts

In this section we focus on the case when both consumer types and products take discrete values. This case can be applied to a variety of industries where the vendors price their products or services based on discrete amounts, and group their consumers with discrete numbers, or even treat each consumer individually. For example, golf clubs often price based on the for number of rounds played, which takes discrete values. Another example is the SaaS provider, TextMarks, that prices its services based on the number of available keywords of each plan. We remark that our paper is the first paper that analyzes 3PT problems with discrete products. Therefore, firms who sell discrete products or services cannot find any guides regarding how to setup an optimal 3PT pricing scheme from the existing literature. Our solution approach of the discrete case not only provide a first framework for firms to follow in order to setup an optimal pricing scheme with discrete products or services, but also pave a way for us to revisit the 3PT problems with continuous products later. Next, we first describe the model and then provide our solution approach. We conclude by an example.

2.1 The model

This model is developed from a vendor's perspective. A monopolistic vendor sells J units of products and assume there are I consumer types. Each consumer type $i, i \in \{i, \ldots, I\}$ has a size α_i. In case the vendor treats each consumer individually rather than putting them into consumer groups, it can be treated as indexing each consumer with an index i and let $\alpha_i = 1$ for each i. The vendor offers a 3PT tariff plan (F, Q, s). If a consumer wants to purchase any units of product, she needs to pay F (access price) first, and can get the first Q units (usage allowance) with no additional charge. If she wants to purchase more than Q units, she needs to pay s for each additional unit. We define F, Q, s as fixed fee, allowance, and overage fee, respectively. Thus, if consumer i purchases j units, she pays $p_j = F$ for $j \leq Q$, or pays $p_j = F + (j - Q)s$ for $j > Q$. Furthermore, we follow Bhargava and Gangwar (2018) to assume zero marginal costs.
Let w_{ij} denote the willingness-to-pay (WTP) of consumer i for j units. Let $\Delta w_{ij} = w_{ij} - w_{i,j-1}$, where $w_{i0} := 0$ be the marginal WTP of consumer i for the jth unit. We follow prior 3PT literature (see Bhargava and Gangwar 2018; Fibich et al. 2017) to assume that the seller has perfect information on consumers' WTPs. In case the seller only knows the stochastic demand, he can use the expected w_{ij} values and the

model will maximize the expected profit for the seller. In fact, the existing literature provide a variety of guidelines regarding how to estimate consumers' WTPs (see, for example, Iyengar and Jedidi 2012; Schlereth et al. 2012; Scholz et al. 2015). In particular, Iyengar and Jedidi (2012) provides a conjoint analysis method that can be used to estimate consumers' WTPs for different quantity amounts of products or services. The seller can directly follow this approach to estimate the w_{ij} values.
We assume that consumers' marginal WTPs are non-negative and weakly decreasing. Furthermore, we assume that consumers' WTP follows Spence-Mirrlees single-crossing property (SCP), which imposes an order to consumer types, such that a higher type consumer (with higher index i value) is always willing to pay more for each unit than a lower type consumer does, and is always willing to pay more for each additional unit. This is a common assumption adopted by nonlinear pricing studies. For example, Bhargava and Gangwar (2018) make the same assumption in their 3PT model with continuous settings. Formally, the following assumptions are adopted on consumers' WTPs:

Assumption 1 (consumers' WTPs).

$$\Delta w_{ij} \geq 0 \qquad \forall i, \forall j, \tag{1}$$

$$\Delta w_{ij} \leq \Delta w_{ij-1} \quad \forall i, \forall j \geq 1, \tag{2}$$

$$w_{ij} \geq w_{i'j} \qquad \forall i > i' \forall j, \tag{3}$$

$$\Delta w_{ij} \geq \Delta w_{i'j} \quad \forall i > i', \forall j. \tag{4}$$

Equations (1) and (2) ensure that consumers' WTPs are non-decreasing and concave. Equations (3) and (4) assume SCP on consumers' WTPs.
Each consumer chooses to purchase a certain number of units $j \in \{1, \ldots, J\}$ that maximizes her surplus $w_{ij} - p_j$. w_{i0} is set to be zero for all i and p_0 is set to be zero, to denote that if consumer i does not purchase any unit, she pays nothing to the seller and gets zero surplus. When a consumer gets maximum surplus from multiple choices of units amount, we assume Without loss of generality that the consumer chooses the units amount that yields the most profit to the seller.
Next, we present our model. Let x_{ij} denote consumers' purchasing decisions, *i.e.*, $x_{ij} = 1$ if consumer i purchases j units and $x_{ij} = 0$ otherwise. Then the 3PT pricing problem for the seller can be formulated as

$$\text{3PTd}: \quad \underset{F,Q,s,p_j,x_{ij}}{\text{Max}} \sum_{i=1}^{I} \sum_{j=0}^{J} \alpha_i x_{ij} p_j$$

$$\text{s.t.} \quad \sum_{j'=0}^{J} (w_{ij'} - p_{j'}) x_{ij'} \geq w_{ij} - p_j \ \forall i, \ \forall j \tag{5}$$

$$\sum_{j=0}^{J} x_{ij} = 1 \ \forall i \tag{6}$$

$$p_0 = 0 \tag{7}$$
$$p_j = F \ \forall j \leq Q \tag{8}$$
$$p_j = F + (j - Q)s \ \forall j > Q \tag{9}$$
$$x_{ij} \in \{0, 1\} \ \forall i, \ \forall j \tag{10}$$
$$F \geq 0, s \geq 0, Q \in \mathbb{Z}^{0+}. \tag{11}$$

Constraint (5) restricts each consumer to maximize her surplus when making purchasing decision. The left hand side models the surplus consumer i receives from her purchasing decision, and the right hand side models her surplus from the purchase of alternate units. Constraint (6) limits each consumer to make one purchasing decision. Constraint (7) assumes the price for zero units to be zero. Constraints (8) and (9) together set up the 3PT prices of each units amount for the given F, Q, and s. Constraint (10) restricts x_{ij} to be binary variables. Constraint (11) restricts F and s to be non-negative, and Q be an integer variable. The objective function captures the total revenue for the vendor. In this model, there are continuous variables F, s, and p_j, integer variable x_{ij} and Q, and nonlinear terms $x_{ij}p_j$. Therefore, it is a Mixed Integer Nonlinear Programming (MINLP) problem, which in general is only solvable for small size problems. Next, we will develop our solution approach that can quickly solve large-sized 3PTd problems. We start by presenting important properties of the optimal solution of 3PTd problem, that will lead to find the solution approach.

2.2 Solution approach

The following properties of the optimal solution will be useful to find the solution approach for 3PTd. The proofs of all the results are provided in the "Appendix".

Proposition 1 *There exists an optimal solution to 3PTd, in which each higher indexed consumer purchases weakly more units than a lower indexed consumer does.*

Proposition 2 *There exists an optimal solution that satisfies* $F^* = w_{i_1 j_1}, i_1 \in \{1, \ldots, I\}, j_1 \in \{1, \ldots, J\}$ *and* $S^* = \Delta w_{i_2 j_2}, i_2 \in \{i_1 + 1, \ldots, I\}, j_2 \in \{j_1 + 1, \ldots, J\}$, *where* F^* *and* s^* *denote the optimal fixed fee and optimal overage fee, respectively. That means, there exists an optimal solution to 3PTd, in which the fixed fee equals to the WTP value of one of the consumers, and the overage fee equals to the marginal WTP value of one of the consumers.*

Proposition 1 is a result of the SCP assumption. This assumption is widely used in non-linear pricing literature to narrow down the search space of optimal consumer allocations to only focus on the ones in which higher type consumers purchase weakly more units than lower type consumers do. Based on this result, we discover Proposition 2, which further narrows down the search space of optimal values of F and s. We find that, for the optimal solution of 3PTd, we only need to consider consumers' WTP values for F and consumers' marginal WTP values s. This result eliminates all the other continuous values from the feasible set of F and s of the optimal solution, which hugely reduces the difficulty level of finding the optimal solution. We remark that, to the best of our knowledge, such a result is not discovered in the literature. One

reason is that, as we presented earlier in the literature review section, previous studies mainly explore the 3PT problem with continuous units, and mainly use first/second order derivatives methods to analyze the optimal solutions. In the continuous space of units, there are still infinitely many consumers' WTP values and consumers' marginal WTP values even with such a result. In 3PTd, the problem is defined with discrete units. Thus, it allows us to discover such a "hidden" property, which leads to a solution approach.

Next, based on Propositions 1 and 2, we propose a dynamic programming (DP) solution approach to solve $3PTd$. According to Proposition 2, the optimal fixed fee F will always be one of the w_{ij} values, and the overage fee S will always be one of the Δw_{ij} values. Next we show that for given F and S values, how to calculate the optimal profit.

When $F = w_{i_1 j_1}, i_1 \in \{1, \ldots, I\}$, $j_1 \in \{1, \ldots, J\}$, and $S = \Delta w_{i_2 j_2}, i_2 \in \{i_1 + 1, \ldots, I\}$, $j_2 \in \{j_1 + 1, \ldots, J\}$, the seller's profit $\Pi(i_1, j_1, i_2, j_2)$ can be calculated as follows:

$$\Pi(i_1, j_1, i_2, j_2) = \sum_{i=i_1}^{I} \alpha_i F + \sum_{i=i_1+1}^{I} \alpha_i (j(i) - j_1) s, \tag{12}$$

where $j(i)$ is defined as $j(i) = \arg\max_j \{j \mid \Delta w_{ij} \geq s, j \geq j(i-1)\}$. The first part in $\Pi(i_1, j_1, i_2, j_2)$ is $\sum_{i=i_1}^{I} \alpha_i F$, which captures the total fixed fees the seller collect. Given $F = w_{i_1 j_1}$, consumers $i \in \{i_1, I\}$ buy non-zero units and thus the seller gets F from each of such consumer types. The second part is $\sum_{i=i_1+1}^{I} \alpha_i (j(i) - j_1) S$, which captures the total overage fees. $j(i)$ is defined to find for each consumer $i \geq i_1 + 1$, the highest unit such that her marginal WTP for this unit is no less than the overage fee S. As such, $j(i)$ also represents the total units that consumer i purchases given F and S. Therefore, each consumer i pays $(j(i) - j_1)S$ for the units exceed the allowance j_1.

Next, we use an example with four consumer types and four units to show how to calculate seller's profit for given $F = w_{i_1 j_1}, i_1 \in \{1, \ldots, I\}$, $j_1 \in \{1, \ldots, J\}$, and $S = \Delta w_{i_2 j_2}, i_2 \in \{i_1 + 1, \ldots, I\}$.

2.3 A numerical example

Example 1 Consider a scenario with 4 units of products and 4 consumer types. The consumer's group size is assumed to be one for each consumer type. Consumers' WTPs and marginal WTPs are given in Table 1.

Note that the w_{ij}'s in Table 1(a) are chosen arbitrarily based on the SCP assumption (see Assumption 1). The Δw_{ij}'s in Table 1(b) are calculated based on the w_{ij} values. In order to find the optimal 3PT pricing scheme for Example 1, according to Proposition 2, we only need to consider w_{ij} values as F^* and Δw_{ij} values as s^*. Next, we show how to calculate seller's profit for given F and s values. We arbitrarily choose $F^* = w_{11} = 14$ and $s^* = \Delta w_{33} = 11$, from Table 1. According to Proposition 2, $F^* = w_{i_1 j_1}$ and $S^* = \Delta w_{i_2 j_2}$, and therefore, $i_1 = 1, j_1 = 1, i_2 = 3, j_2 = 3$. Thus, we have

Table 1 w_{ij} and Δw_{ij} values for Example 1

(a) Units	w_{ij}			
	I_1	I_2	I_3	I_4
0	0	0	0	0
1	14	18	30	60
2	24	30	45	90
3	29	38	56	110
4	30	40	60	120
(b) Units	Δw_{ij}			
	I_1	I_2	I_3	I_4
0	0	0	0	0
1	14	18	30	60
2	10	12	15	30
3	5	8	11	20
4	1	2	4	10

$j(2) = \arg\max_j\{j \mid \Delta w_{2j} \geq 11, j \geq 1\} = 2$. This is because, in Table 1(b), the maximum j value where $\Delta w_{2j} \geq 11$ is 2, since $12 \geq 11$. $j(3) = \arg\max_j\{j \mid \Delta w_{3j} \geq 11, j \geq 2\} = 3$, and $j(4) = \arg\max_j\{j \mid \Delta w_{4j} \geq 11, j \geq 3\} = 3$. Note that $j(1) = 1$ as $i_1 = 1$ and $j_1 = 1$ (this means that consumer i_1 is the first one who is assigned non-zero units). Following Equation (12), we have $\Pi(1, 1, 3, 3) = \sum_{i=1}^{4} 14 + \sum_{i=2}^{4}(j(i) - 1)11 = 111$. In this solution, when seller charges \$14 as fixed fee with one unit allowance, and \$11 as overage fee for each additional unit, all four consumers purchase the plan. Consumer 1 pays \$14 to get the allowance unit. Consumer 2 gets two units in total with a payment of \$25. Both consumers 3 and 4 get three units in total with a payment of \$36. It leads to a total profit of \$111 for the seller.

Now we describe how to get the global optimal solution. In Table 2, we list the maximum profits for each feasible combination of Q, F, and s values. For example, when $Q = 1$, $F = 14$, and $s = 11$, the maximum profit is 111. The detailed calculation of this case is given in the previous paragraph. We basically use Equation (12) to calculate the profits for each possible situation. After calculating each profit value, the maximum value in this table yields the global optimal solution. For Example 1, the optimal solution is $Q^* = 1$, $F^* = 24$, and $s^* = 8$. The optimal value is 128. Note that the blank cells in Table 2 indicate infeasible solutions. Since the cost to calculate each $\Pi(i_1, j_1, i_2, j_2)$ in Equation (12) is $O(I)$, the total cost for the whole approach to find the global optimal solution is $O(I^3 J^2)$.

Now we consider the case, which the seller adopts a transitional pricing strategy—unit pricing—in this example. In this case, the seller sets a price for each unit and each consumer keeps purchasing until her marginal WTP is less than the unit price. We can easily find that the optimal unit price for this example is 10. Consumers 1, 2, 3, and 4 purchase 1, 2, 3, and 4 units, respectively. The seller earns 100 as the total profit.

Table 2 Optimal Profits for each Possible Q, F, and s for Example 1

Q	F	s									
		0	12	8	2	15	11	4	30	20	10
1	14		104	112	74	101	111	88	86	96	116
1	24			128∗	108		118	116		116	126
1	29				122			124			126
1	30	120									
2	18					99	98	78	84	94	104
2	30						112	106		110	120
2	38							122			124
2	40	120									
3	30								90	100	90
3	45									110	110
3	56										121
3	60	120									
4	60	60									
4	90	90									
4	110	110									
4	120	120									

Comparing the profit of unit pricing with that 3PT, it turns out that 3PT increases the profit by 28% for the seller.

3 Discrete consumer types and continuous amounts

In Sect. 2, we solved the 3PT problem with discrete consumer types and discrete units. Now we analyze the case where the discrete units become a continuous amount, while the consumer types are still discrete. Such a model fits better with firms that sell continuous amounts of products or services, such as telecommunication service providers who sell data amounts. This model is studied in Fibich et al. (2017) with two consumer types. We allow the number of consumer types to be any number.

We will discretize the continuous amounts so that we can use the discrete solution approach to handle the problem. We remark that the standard piecewise linear approximations approaches cannot directly be applied to discretize this problem, since this problem involves consumer self-selection. If we use the standard piecewise linear approximation approach to solve the problem, each consumer is not guaranteed to choose the desired amount with the obtained approximated prices. Next, we first introduce the model with continuous units. Then we present two important results: (1) With piecewise linear approximation, all the optimal allocations always happen on the kink points; and (2) Each consumer always chooses the desired amount generated from the piecewise linear approximation. With these two results, we extend our discrete solution to solve the model with continuous amounts.

3.1 The model

This model is largely similar to the one we studied in the previous section. The major difference is that, the discrete unit variable j in 3PTd now becomes a continuous variable $y \in \Re_+$. Consequently, we let $p(y)$ denote the price consumer pays for y amount and $w_i(y)$ denote consumer type i's WTP for y amount. The definitions of F, Q, s, and $\alpha(i)$ remain the same as in 3PTd. We continue to assume that consumers' WTPs are non-decreasing and concave, and follow the SCP as follows:

Assumption 2 (Consumers' WTPs with continuous amounts).

$$w_i'(y) \geq 0 \ \forall i, \qquad \forall y, \tag{13}$$
$$w_i''(y) \leq 0 \ \forall i, \qquad \forall y, \tag{14}$$
$$w_i(y) \geq w_{i'}(y) \ \forall i > i', \quad \forall y, \tag{15}$$
$$w_i'(y) \geq \Delta w_{i'}'(y) \ \forall i > i', \ \forall y. \tag{16}$$

Define $y_i, i \in \{1, \ldots, I\}$ to denote the amount consumer type i purchases. Then, the seller's decision problem is defined as follows:

$$\text{3PTc}: \ \underset{y_i, p(y)}{\text{Max}} \sum_{i=1}^{I} \alpha_i \, p(y_i)$$
$$\text{s.t.} w_i(y_i) - p(y_i) \geq w_i(y) - p(y) \ \forall i \ \forall y \tag{17}$$
$$p(0) = 0 \tag{18}$$
$$p(y) = F \ \forall y \leq Q \tag{19}$$
$$p(y) = F + (y - Q)s \ \forall Q \leq y \leq Y \ (11). \tag{20}$$

Constraints (17), (18), (19), and (20) can directly be mapped to Constraints (5), (7), (8), and (9), respectively in 3PTd. Constraint (6) is not needed anymore, because in 3PTc, we use y_i to denote the amount purchased by consumer i, which guarantees that each consumer makes one purchasing decision. The objective function can also be mapped to that in the discrete case. In general, 3PTc model is not solvable with any existing solvers, since it contains a function $p(y_i)$ as a variable. Next, we develop a solution approach for 3PTc problem, that can generate an approximation value within any given error tolerance to its pure optimal value.

3.2 Solution approach

Proposition 1 that was proved in Sect. 2 with discrete units remain true with continuous amounts. Thus we have the following property.

Proposition 3 *There exists an optimal solution to 3PTc that satisfies:*

$$y_i^* \leq y_{i+1}^* \ \forall i \leq I - 1. \tag{21}$$

Next, we develop a solution approach for 3PTc. We first investigate a special case of 3PTc problem, where the WTP functions for each consumer type i, $w_i(y)$, is piecewise linear. We will show that such a problem can be solved as a discrete problem we treated in Sect. 2. Then we show that we can solve any 3PTc problem with general continuous $w_i(y)$ functions by using its piecewise linear approximation functions. The obtained piecewise linear approximation solution has an objective value that is within any given error tolerance to the pure optimal objective value.
First, we show that, for a 3PTc problem, when its $w_i(y)$ functions are piecewise linear, the solution can be solved as a discrete 3PTd problem. Let $w_i(\cdot)$ be piecewise linear functions whose breakpoints form a subset of $\{k_0, k_1, \ldots, k_J\}$, such that for $i \in \{1, \ldots, J\}$, it satisfies $0 = k_0 < k_1 < \ldots < k_J = Y$. For this case, we find the following result:

Proposition 4 *When $w_i(\cdot)$ are piecewise linear functions with breakpoints $\{k_1, \ldots, k_J\}$, there always exists an optimal solution for 3PTc, such that for each i, y_i^* lies to one of the breakpoints, i.e., $y_i^* \in \{k_0, \ldots k_J\}$ $\forall i$.*

With such a result, we can always find an optimal solution for a piecewise linear 3PTc problem such that all its y_i^* values lie to the breakpoints. Therefore, if we just keep all the breakpoints values of each $w_i(y)$ function and remove all the interior lines that connect the breakpoints, we will not change the optimal solution of the problem. Given that, we can solve any piecewise linear 3PTc problem as the discrete 3PTd which we treated in the previous section.

Theorem 1 *When $w_i(\cdot)$ are piecewise linear functions with breakpoints $\{k_1, \ldots, k_J\}$, the 3PTc problem can be solved as the following discrete problem 3PTxl:*

$$3PTxl: \underset{F,Q,s,p(k_j),x_{ij}}{Max} \sum_{i=1}^{I}\sum_{j=0}^{J} \alpha_i x_{ij} p(k_j)$$

$$s.t. \sum_{j'=0}^{J} (w_i(k'_j)) - p(k'_j))x_{ij'} \geq w_i(k_j) - p(k_j) \ \forall i, \ \forall j \tag{22}$$

$$\sum_{j=0}^{J} x_{ij} = 1 \ \forall i \tag{23}$$

$$p(0) = 0 \tag{24}$$

$$p(k_j) = F \ \forall k_j \leq Q \tag{25}$$

$$p(k_j) = F + (j - Q)s \ \forall k_j > Q \tag{26}$$

$$x_{ij} \in \{0, 1\} \ \forall i, \ \forall j. \tag{27}$$

Now we consider the general case where $w_i(\cdot)$ are Lipschitz continuous functions with Lipschitz constant β, which means that we assume $w_i(y_1) - w_i(y_2) \leq \beta y_1 - y_2, \ \forall y_1 \in [0, Y], \ \forall y_2 \in [0, Y]$. We use the previous result to solve the general 3PTc problem by constructing piecewise linear approximation for $w_i(y)$ as follows. Let ϵ be a given error tolerance. Then, we choose $k = \frac{\epsilon}{\beta}$ and $J = \lceil \frac{Y}{k} \rceil$. Let $k_j = jk$ for $j \in \{0, \ldots, J-1\}$

and $k_J = Y$. Then, for each i, we define a piecewise linear function $w_i^k(y)$ as

$$w_i^k(y) = \frac{k_{j+1} - y}{k_{j+1} - k_j} w_i(k_j) + \frac{y - k_j}{k_{j+1} - k_j} w_i(k_{j+1}) \text{ for } k_j \le y \le k_{j+1} \tag{28}$$

Let Π^k denote the optimal profit when $w_i^k(y)$ is the WTP for each consumer type i and Π^c denote the optimal profit of 3PTc.

Theorem 2 *For a given error tolerance ϵ, define $k = \frac{\epsilon}{\beta}$. Then, $\Pi^k - \epsilon \le \Pi^c \le \Pi^k + \epsilon$.*

In Theorem 2, we show that for any given error tolerance, we can choose a corresponding grid step to piecewise-linearize the WTP functions of a 3PTc problem and convert it to a piecewise linear problem 3PTxl. Then we use the solution approach of the discrete model to solve 3PTxl. The obtained optimal solution is guaranteed to bound the optimal solution of the original continuous problem 3PTc within the given error tolerance.

Next, we use an example to demonstrate how to solve a 3PTc problem with general continuous WTP functions. We will also show how the gap between the solution of the piecewise linear approximation and the true optimal value converges with choices of the grid step.

3.3 A numerical example

Example 2 Consider a scenario with four consumer types with group size 1 for each type. The seller sells a continuous amount of units, y, with maximum amount of 4 units. Each consumer $i, i \in \{1, 2, 3, 4\}$ has a willingness-to-pay function as $w_i(y) = (0.9 + 0.1i)y - \frac{y^2}{8}$.

We use a global optimization solver, BARON (Tawarmalani and Sahinidis 2002), to solve Example 2 with the 3PTc formulation. The optimal solution found is to set the three part tariff pricing scheme with $F = 1.955$, $Q = 3.4$, and $s = 0.2$. Given such a pricing scheme, consumers 1, 2, 3, and 4 buy units 3.4, 3.6, 4, and 4 respectively. The sellers earns 8.1 as the total profit.

Next we demonstrate to solve this example by using our solution approach. We start with and error tolerance $\epsilon = 1$, which leads to a grid step $k = \frac{\epsilon}{\beta} \approx 1$ (the β of the functions given in this example is 1.3). Using this grid step, we generate a 3PTxl problem with $w_i(k_j)$ and $\Delta w_i(k_j)$ values as shown in Table 3.

Then we use the solution approach we developed for 3PTd in Sect. 2.2 to quickly solve this problem. The approximation optimal solution is to set $F = 1.875$. $Q = 3$, and $s = 0.225$. With such a pricing scheme, consumers 1, 2, 3, and 4 buy units 3, 4, 4, and 4 respectively. The seller earns 8.175 as total profit. The actual error (the gap between our approximation and the true optimal solution) is $8.1 - 8.175 = 0.075$, which is smaller than the error tolerance, 1.

We can further reduce the grid step value to reduce the gap between our approximation and the true optimal solution, as shown in Table 4. The second column given in the table shows the true optimal solution. The third through the last columns show the

Table 3 $w_i(k_j)$ and $\Delta w_i(k_j)$ values for Example 2 with grid step 1

(a) Units	$w_i(k_j)$			
	I_1	I_2	I_3	I_4
k_0	0	0	0	0
k_1	0.875	0.975	1.075	1.175
k_2	1.5	1.7	1.9	2.1
k_3	1.875	2.175	2.475	2.775
k_4	2	2.4	2.8	3.2
(b) Units	$\Delta w_i(k_j)$			
	I_1	I_2	I_3	I_4
k_0	0	0	0	0
k_1	0.875	0.975	1.075	1.175
k_2	0.625	0.725	0.825	0.925
k_3	0.375	0.475	0.575	0.675
k_4	0.125	0.225	0.325	0.425

Table 4 Approximation solutions with reduced grid step for Example 2

Grid Step	True solution	Approximation			
		1	0.1	0.01	0.001
F	1.955	1.875	1.955	1.955	1.955
Q	3.4	3	3.4	3.4	3.4
s	0.2	0.225	0.212	0.2013	0.2001
Profit	8.1	8.175	8.1175	8.1017	8.1002
Actual Error		0.075	0.0175	0.0017	0.0002

approximation solution for grid steps 1, 0.1, 0.01, and 0.001 respectively. In the last row, it shows that the gap between the approximation and the true optimal solution converges to zero as the grid step decreases.

We remark that, in general, 3PTc problem is a nonlinear optimization problem, and is not guaranteed to be solvable by using any existing solver. We are able to obtain the optimal solution for this specific example mainly because the problem size is small (with four consumer types), and the quadratic willingness-to-pay functions are easily to handle. When the problem size become large, or the willingness-to-pay functions become hard to handle, it is not guaranteed that the global optimization solver is still able to converge to the optimal solution. On the other hand, our proposed solution can solve large scale problems with any type of willingness-to-pay functions. We can choose any error tolerance and use our solution approach to obtain a solution for which the actual error is guaranteed to be within the given error tolerance.

4 Continuous consumer types and continuous amounts

In this section, we further extend our model to study more general three part tariff pricing problems, in which not only the amounts are continuous, but also the consumer types. We remark that, this model is the same as the problem studied in Bhargava and Gangwar (2018). In their paper, the authors provide some properties for its optimal solutions. However, the authors claimed that "The 3PT optimization problem with heterogeneous consumers is generally intractable.", and do not provide an optimal solution approach for the problem. We will present a solution approach for this problem, which generates an approximation value that can be within any error tolerance to the pure optimal value.

4.1 The model

Let consumer types be indexed with $\theta \in [\underline{\theta}, \bar{\theta}]$ with density distribution function $f(\theta)$ and cumulative distribution function $F(\theta)$. Let $w(\theta, y)$ denote consumer type θ's WTP for y amount. The definitions of F, Q, s, and $y(p)$ remain the same as those in Sect. 3. Consumers' WTPs are still assumed to be concave and non-decreasing, and follow SCP:

Assumption 3 (Consumers' WTPs with continuous amounts and continuous consumer types)

$$\frac{\partial w(\theta, y)}{\partial y} \geq 0 \ \forall \theta, \ \forall y, \tag{29}$$

$$\frac{\partial^2 w(\theta, y)}{\partial y^2} \leq 0 \ \forall \theta, \ \forall y, \tag{30}$$

$$\frac{\partial w(\theta, y)}{\partial \theta} \geq 0 \ \forall \theta, \ \forall y, \tag{31}$$

$$\frac{\partial^2 w(\theta, y)}{\partial \theta \partial y} \geq 0 \ \forall \theta, \ \forall y. \tag{32}$$

Define $y(\theta)$ to denote the amount consumer θ purchases. The seller's decision problem is as follows:

$$\text{3PTg}: \underset{F, Q, s, y(\theta), p(y)}{\text{Max}} \int_{\underline{\theta}}^{\bar{\theta}} f(\theta) p(y(\theta)) d\theta$$

$$\text{s.t.} w(\theta, y(\theta)) - p(y(\theta)) \geq w(\theta, y) - p(y) \ \forall \theta \ \forall y. \tag{33}$$

$$p(0) = 0 \tag{34}$$

$$p(y) = F \ \forall y \leq Q \tag{35}$$

$$p(y) = F + (y - Q)s \ \forall Q \leq y \leq Y \tag{36}$$

In order to solve 3PTg, we need the following result to discretize the continuous consumer types.

4.2 Solution approach

Proposition 5 *For any given interval* $(\theta^l, \theta^h]$ *where* $\underline{\theta} \leq \theta^l \leq \theta^h \leq \bar{\theta}$*, we achieve an underestimation (respectively overestimation) of the optimal value of 3PTg by replacing* $w(\theta, y)$ *of all consumer types* $\theta \in (\theta^l, \theta^h]$ *with* $w(\theta^l, y)$ *(respectively* $w(\theta^h, y)$*).*

With this result, we develop the following solution approach for 3PTg, such that the gap between the optimal value of the proposed solution and the true optimal value is within any given error tolerance. Let m be a grid step length for θ, and $I = \lceil \frac{\bar{\theta}-\underline{\theta}}{m} \rceil$. Let $m_i = \underline{\theta} + im$ for $i \in 0, \ldots, I-1$ and $m_I = \bar{\theta}$. Let $w^m(\theta, y) = w(m_i, y)$ for $m_i \leq \theta < m_{i+1}$. By underestimating $w^m(\theta, y)$ in this way, we discretize a 3PTg problem on θ to a 3PTc problem we treated in Sect. 3. We further discretize this problem on y by choosing $k = \frac{\epsilon}{2\beta}$, $J = \lceil \frac{Y}{k} \rceil$, and $k_j = jk$ for $j \in 0, \ldots, J-1$ and $k_J = Y$. Eventually, we discretize $w(\theta, y)$ on the continuous (θ, y) space to w^d_{ij} on the discrete (i, j) space by letting: $w^d_{ij} = w(m_i, k_j)$ for $m_i \leq \theta < m_{i+1}$ and $k_j \leq y < k_{j+1}$. Let Π^g be the optimal value of a 3PTg problem with $w(\theta, y)$, and Π^{gd} be the optimal value of a 3PTd problem with corresponding w^d_{ij}.

Theorem 3 *For a given error tolerance* ϵ*, define* $m = \frac{\epsilon}{2\beta}$ *and* $k = \frac{\epsilon}{2\beta}$*. Then,* $\Pi^{gd} - \epsilon \leq \Pi^g \leq \Pi^{gd} + \epsilon$.

Theorem 3 shows that we can always find an interval size on the consumers types to discretize the continuous consumers. The true optimal solution of the continuous problem is bounded by the solution of the discretized problem with any given error tolerance. Next, similar as previous sections, we use an example to demonstrate how this solution approach works with decreasing error tolerances. We further provide some additional computational experiments to demonstrate that the improvements achieved with our solution approach reduce quickly with the decreasing grid steps.

4.3 Numerical results

Example 3 Consider a scenario with a consumer group. Consumers' type θ is uniformly distributed between 1 and 5. The seller sells a continuous amount of units, y, with maximum amount of 4 units. Each consumer θ, $\theta \in [1, 5]$ has a willingness-to-pay function as $w(\theta, y) = (0.9 + 0.1\theta)y - \frac{y^2}{8}$.

First, we remark that we cannot find the pure optimal solution for this problem, since there is no solution approach available in the literature for 3PTg. We will illustrate how our approximation solutions converge to the pure optimal solution with decreasing grid steps.
We start with a grid step $k = 1$ and $m = 1$, which generates a 3PTxl with the same $w_i(k_j)$ and $\Delta w_i(k_j)$ values as shown in Table 3 for Example 2. The only difference is that the group size for each consumer is $\frac{1}{4}$ here, instead of being 1 in Example 2. By using the same solution approach, we get the optimal value of this case to be 2.044.

Table 5 Approximations with reduced grid steps for Example

Grid Step	1	0.2	0.1	0.02
Approximations	2.0438	2.0456	2.0466	2.0471

Table 6 Approximations with reduced grid steps for five randomized $w(\theta, y)$ functions

	Grid Step			
Experiment Index	1	0.2	0.1	0.02
1	5.5132	5.5147	5.5151	5.5152
2	2.6701	2.6730	2.6742	2.6747
3	5.2915	5.2921	5.2922	5.2922
4	7.1198	7.1223	7.1229	7.1232
5	3.9190	3.9224	3.9235	3.9239

In order to get more accurate approximations for the optimal solution, we can keep decreasing the grid steps. Table 5 shows the optimal values with grid steps of both k and m to be 1, 0.2, 0.1, and 0.02 respectively. In this example, we observe that approximation values get improved with decreasing grid steps. However, such improvements decrease quickly with the decreasing grid steps. Therefore, our solution approach could reach a reasonably good approximation result with a relatively large grid step, which means that we can achieve a reasonably good result with a relative low computational cost.

We further test our proposed solution approach with general functions $w(\theta, y) = ay + b\theta + cy\theta - dy^2 - e\theta$, where $\{a, b, c, d, e\}$ are parameters that are taken random values such that $w(\theta, y)$ satisfy Assumptions (29)–(32). We show five of the results in Table 6. These results confirm that our proposed solution approach yields reasonably good results even when the grid steps are relatively large. Therefore, we can easily control the computational costs of our solution approach.

4.4 Comparison to results in prior work

As remarked earlier, the 3PTg is same as the 3PT problem studied in Bhargava and Gangwar (2018). Both models have continuous consumer types and amounts, and both require consumers' WTPs to follow single-crossing property. One of the main results in Bhargava and Gangwar (2018) regarding the solution approach of the 3PT is that they develop a reformulation of the problem given in their Corollary 1. The reformulation simplifies the original 3PT model, which is considered as "intractable" in their paper. Using this reformulation, they provide a solution approach for the problem with discrete consumer types and continuous amounts. When consumer types are continuous, they solve the problems with a specific consumer's WTP function: $w(\theta, y) = \theta y - \frac{\beta}{2} y^2$, where θ and y are the consumer index and amount index, respectively. In addition, they require the consumer types to follow one of the three distributions: heterogeneous uniform distribution, left-skewed triangle distribution, or symmetric unimodal distribution. Guidelines regarding how to solve the reformulation in general beyond the specific WTP function and the three specific distribution

functions are not provided. On the other hand, the solution approach presented in the current study allows any type of functions for consumers' WTPs, as far as the single-crossing property is satisfied. In addition, we do not impose any restrictions on the type of consumer distributions. Therefore, our solution approach significantly expands the scope of continuous 3PT problems that can be solved.

Moreover, we would like to emphasize that, once one cannot obtain closed-form solutions from the reformulation in Bhargava and Gangwar (2018), then the approximation solutions obtained by using standard discretization methods on their reformulation could be arbitrarily far from the real optimal solution. This is because the 3PT involves consumer self selection. The discretization solution is generated by using the discretized consumer's WTP functions. When the pricing strategies are adopted, consumers could make different purchase decisions since their real continuous WTPs are different than the discretized WTPs. Therefore, when one uses standard discretization approximation methods to solve for formulation (4) in Bhargava and Gangwar (2018), the gap between the approximation and the real optimal value cannot be bounded. In our Theorems 2 and 3, we develop specific ways of discretizations, such that consumers' purchasing decisions do not change when the continuous units and continuous consumers types are discretized. With these results, we guarantee that the gap between our approximation and the real solution is bounded by the discretization grid steps, so that we can always find a grid step to obtain an approximation that is within any given error tolerance. These results are indeed among the main contributions of our paper.

5 Conclusion

This study analyzes the 3PT pricing problem, in which service providers charge a fixed fee with an allowance of free units, and a per-unit fee for additional units above the allowance. We assume that consumer's willingness to pay for different units follow Spence-Mirrlees single-crossing property. An innovative solution approach is presented for the case when both the consumer types and the amounts are discrete, which is novel in the literature. The cost of the solution is polynomial to both the number of units and the number of consumer types. We then extend our analysis to the case of discrete consumer types and continuous amounts. Unlike other studies, our solution approach does not put any simplifying assumption on the type of the consumer. Finally, we expand our analysis to the case of continuous consumer types and amounts. We present solution approaches for both of these two cases. Our solutions provide approximations within any given error tolerance to the true optimal value.

Before our paper, the existing solution approaches can only be used to solve 3PT problems under certain restricted conditions. This paper provides an analytical solution approach that significantly expands the scope of 3TP problems that can be solved efficiently. The major contribution of this paper is to provide a framework that guides firms to obtain the optimal pricing scheme if they are interested in adopting 3PT to sell their products or services.

One future extension area of this research is to relax the Spence-Mirrlees single-crossing property as an assumption for the consumer's willingness-to-pays. Another fruitful area of future research is to consider a duopoly or an oligopoly market for the

sellers, since so far all the theoretical studies on 3PT focus on the monopoly setting. Other possible areas of research is to include different kinds of sellers' costs, such as fixed costs or marginal production costs, into the 3PT model.

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Declarations

Conflict of interest The authors have no competing interests to declare that are relevant to the content of this article.

Appendix A: Proof of Proposition 1

Proof Let $[F', Q', s', x'_{ij}]$ be an optimal solution of $3PTd$. Let $i = \arg\min_i\{i \mid j > j', x'_{ij} = x'_{i+1j'} = 1\}$. This means that i is the lowest type consumer who purchases strictly more units than consumer type $i+1$ does. Consider the following chain of inequalities:

$$w_{ij} - w_{ij'} \geq p'_j - p'_{j'} \geq w_{i+1,j} - w_{i+1j'} \geq w_{ij} - w_{ij'},$$

where the first and second inequalities follow because consumer type i purchases j and consumer type $i+1$ purchases j', respectively, and the third inequality follows SCP assumption. Therefore, equality holds throughout. It follows that $w_{i+1,j} - p'_j = w_{i+1j'} - p'_{j'}$, meaning that the seller can allocate consumer type $i+1$ to purchase j units instead of j' units. Since $j > j'$, the new allocation does not decrease the seller's profit. With such a procedure, we get a new optimal solution in which consumer type i no longer purchases more units than consumer type $i+1$ does. Since there are finitely many consumer types, we can repeat the same procedure in finitely many times to get an optimal solution, in which there does not exist any consumer type i who purchases strictly more units than consumer type $i+1$ does. □

Appendix B: Proof of Proposition 2

Proof We first show that, for any given solution $[F', Q', s']$, we can always find another solution with fixed fee F''. The new solution yields no less profit to the seller, and satisfies $F'' \in \{w_{11}, \ldots, w_{IJ}\}\ \forall i,\ \forall j$.
When the price scheme is $[F', Q', s']$, consumers can choose to purchase $j \in \{Q', Q'+1, \ldots, J\}$ units with price $p_j = F' + (j - Q')s'$ for each j. Given such pricing scheme, let $\{j_1, j_2, \ldots, j_K\}$ be the amounts that are purchased by at least one consumer type. Let i_1 and i_2 be the lowest indexed consumer and highest indexed consumer who purchase j_1, respectively. According to Proposition 1, each consumer $\{i_1, i_1+1, \ldots, i_2\}$ purchases j_1, and each consumer $\{i_2+1, i_2+2, \ldots, I\}$ purchases a quantity in $\{j_2, j_3, \ldots, j_K\}$.

Now we define the new solution as follows. Let Q'' be the actual units amount purchased by consumer i_1, *i.e.*, $Q'' = j_1$, and F'' be her WTP for Q'' units, *i.e.*, $F'' = w_{i_1 j_1}$. By definition, F'' in the new solution equals to the WTP of one of the consumers. We next show that the solution $[F'', Q'', s']$ yields no less profit to the seller. Since consumer i_1 purchases Q'', by Proposition 1, consumers $\{i_1, i_1+1, \ldots, I\}$ purchase at least Q'' units. Since s' does not change, each consumer type still stops purchasing at the unit for which her marginal WTP is lower than s'. Thus each consumer $\{i_1, i_1 + 1, \ldots, I\}$ still purchases the same amounts of units. In addition, $F'' = w_{i_1 j_1} \geq w_{i_1 Q'} \geq F'$, where the first equality is by definition of F'', first inequality by $j_1 \geq Q'$, and second inequality by consumer i_1 purchases Q' units at price F'. As a result, seller's profit from fixed fee increases by $\sum_{i=i_1}^{I} \alpha_i (F'' - F')$, and his profit from overage fee remains the same, which leads to a non-decreasing total profit.

So far we proved that there exists an optimal solution that satisfies $F^* \in \{w_{11}, \ldots, w_{IJ}\}\ \forall i,\ \forall j$. Now we prove that for such an optimal solution, $s^* \in \{\Delta w_{11}, \ldots, \Delta w_{IJ}\}\ \forall i,\ \forall j$. Consider an optimal solution $[F', Q', s']$, which satisfies $F' \in \{w_{11}, \ldots, w_{IJ}\}\ \forall i,\ \forall j$ and $s' \notin \{\Delta w_{11}, \ldots, \Delta w_{IJ}\}\ \forall i,\ \forall j$. We next construct another solution with s'', such that $s'' \in \{\Delta w_{11}, \ldots, \Delta w_{IJ}\}\ \forall i,\ \forall j$ and it yields no less seller's profit than the original solution.

Since now F' takes one of the values in $\{w_{11}, \ldots, w_{IJ}\}\ \forall i,\ \forall j$, let $w_{i_1 j_1}$ be such value, *i.e.*, $w_{i_1 j_1} = F'$. Let $s'' = \arg\min\{\Delta w_{ij} \mid \Delta w_{ij} > s', i \geq i_1, j \geq j_1\}$. It means that we check every marginal WTP value of consumers who purchase non-zero units, and choose the lowest one that is greater than s'. Obviously, each consumer will remain to purchase the same number of units. This is because, given our procedure of generating F', each consumer $i' \in \{i_1, i_{i+1}, \ldots, I\}$ always purchases j' units that satisfies $\Delta_{i'j'} \geq s''$ and $\Delta_{i'j'+1} \leq s''$. Given the way we are constructing s'', j' will not change for each i'. Since $s'' > s'$, such a new solution will never decrease seller's profits. □

Appendix C: Proof of Proposition 4

Proof Suppose that for a piecewise linear 3PTc problem, there exists an optimal solution $[F', Q', s']$ such that for some i, $y_i^* \notin \{k_0, \ldots, k_J\}$. We prove this proposition by showing that we can always construct a new solution that yields no less profit, and for each i, it satisfies that $y_i^* \in \{k_0, \ldots, k_J\}$.

Let us first assume the optimal allowance amount Q' is on one of the breakpoints, and thus, we have $Q' = k_j$. In this case, it is straightforward to see that there must exist an optimal solution, in which $y_i^* \in \{k_0, \ldots, k_J\}$. This is because, for any consumer i who purchases no less than k_j amount, she will continue to purchase more amount if her marginal WTP is no less than the overage fee, *i.e.*, it satisfies that $w_i'(y) \geq s'$. She stops to purchase more once $w_i'(y) < s'$. Since $w_i'(y)$ is concave piecewise linear, $w_i'(y)$ is a constant for any interior part other than the breakpoints. So consumer i purchases $k_{j'}$ amount that satisfies $w_i'(k_{j'}) \geq s' > w_i'(k_{j'})$.

Now we consider the case when Q' is not on any breakpoints. So, let $k_j' < Q' < k_{j+1}'$. By the same argument that any consumer i who purchases more than zero amount

always continues to purchase more when $w_i'(y) \geq s'$, it is easy to see that y_i^* cannot be on any non-breakpoints other than $y_i^* = Q'$. In other words, y_i^* can either be on one of the breakpoints, or $y_i^* = Q'$. Let $i_1 = \arg\min_i\{i \mid y_i^* = Q'\}$ and $i_2 = \arg\max_i\{i \mid y_i^* = Q'\}$. *i.e.*, let i_1 and i_2 be the lowest and highest consumer types who purchase Q'. According to Proposition 1, any consumer type $i \in \{i_1, \ldots, i_2\}$ purchases Q', and any consumer $i \in \{i_2+1, \ldots, I\}$ purchases an amount that belongs to $\{k'_{j+1}, \ldots, k_J\}$. Now we construct a new solution as follows. Let the allowance Q'' be k'_{j+1}, the fixed fee F'' be $w_{i_1}(k'_{j+1})$, and the overage fee remains the same. In this new solution, consumers $i \in \{i_1, \ldots, i_2\}$ will purchase at least Q'', because they get non-negative surplus from Q''. As a result, the sellers' profit from consumers $i \in \{i_1, \ldots, i_2\}$ does not decrease, since $F'' = w_{i_1}(k'_{j+1}) \geq w_{i_1}(Q') \geq F'$. The equality is by definition of F'', the first inequality is by non-decreasing $w_{i_1}(\cdot)$, and the second inequality is by consumer i_1 purchases Q' at price F' in the original solution. The sellers' profit from consumers $i \in \{i_2 + 1, \ldots, I\}$ does not change. This is because the amounts purchased by these consumers remain the same, as the overage fee remains the same. Therefore, the seller's overall profit does not decrease with the constructed solution, in which $y_i^* \in \{k_0, \ldots, k_J\}$ for each i. □

Appendix D: Proof of Theorem 2

Proof We first show that $\Pi^k - \epsilon \leq \Pi^c$. Let $w_{ij} = w_i^k(k_j)$ to convert 3PTxl to 3PTd. Then we solve 3PTd to find Π^k and the piecewise linear approximation optimal solution Q_1, F_1, s_1 and p_j. Denote $p^k(k_i)$ be the price each consumer i pays in 3PTxl. Now let this approximation solution Q_1, F_1, s_1 be implemented in the original problem 3PTc with $w_i(y)$. Let $p^c(k_i)$ be the price paid in 3PTc and $\Pi^{c'}$ be the obtained profit. We consider each consumer i_1 who purchases weakly more than Q_1 units in 3PTd, and let k_{j1} be the units purchased by i_1. We will show that, the price the seller collects from consumer i_2 in 3PTc is close to that in 3PTxl, *i.e.*, $p^c(k_{i2})$ is close to $p^k(k_{i2})$. First, we argue that consumer i_1 will still purchase weakly more than Q_1 units in 3PTc. This is because her WTP on Q_1 is no less than F_1, *i.e.*, $w_{i_1}(Q_1) = w_{i_1}^k(Q_1) \geq F_1$. The equality is because of Proposition 4, which suggests that Q_1 lies to one of the breakpoints in 3PTxl. The inequality is because consumer i_1 purchases weakly more than Q_1 units in 3PTd. Second, in case k_{j1} is strictly more than Q_1 in 3PTxl, we argue that, in 3PTc, consumer i_1 will purchase k_{j2} units, which satisfies $k_{j2} \geq k_{j1} - k$, *i.e.*, it is possible that consumer i_1 purchases fewer units than she did in 3PTxl, but the difference is no more than k units. This is because $w_{i1}'(k_{j1} - k) \geq \frac{w_{i1}(k_{j1}) - w_{i1}(k_{j1}-k)}{k} \geq s_1$. The first inequality is by the concavity assumption on $w_i(\cdot)$. The second inequality is by consumer i_1 purchases k_{j1} units in 3PTxl so that her marginal WTP $\frac{w_{i1}(k_{j1}) - w_{i1}(k_{j1}-k)}{k}$ is weakly more than the overage fee s_1. Since the marginal WTP of consumer i_2 on $k_{j1} - k$ units is no less than the overage fee s_1, she buys at least $k_{j1} - k$ units. Therefore, we have

$$p^k(k_{i2}) - p^c(k_{i2}) \leq ks_1 \leq kw_I'(0) \leq k\beta. \tag{D1}$$

The second inequality is because the maximum possible value of s_1 is $w_I'(0)$. The last inequality is by that β is the Lipschitz constant of $w_I(\cdot)$. Given such results, we calculate the gap between Π^k and Π^c as follows:

$$\Pi^k - \Pi^c \leq \Pi^k - \Pi^{c'} = \sum_{i=1}^{I} \alpha_i (p^k(k_i) - p^c(k_i)) \leq k\beta \sum_{i=1}^{I} \alpha_i = k\beta = \epsilon.$$

The first inequality is because Π^c is the optimal value of 3PTc while $\Pi^{c'}$ is the objective value of one of its feasible solutions and the second is by (D1). The first equality is by the definitions of Π^k and $\Pi^{c'}$, the second is by $\sum_{i=1}^{I} \alpha_i$, which is normalized to one, and the last is by the definition of ϵ.

Next, we prove $\Pi^c \leq \Pi^k + \epsilon$. Let Q_2, F_2, s_2 be an optimal solution of a given 3PTc problem. Let y_i' be the units each consumer i purchases in this solution and $p^c(y_i)$ be the corresponding price. Let i' be the lowest consumer type who purchases weakly more than Q_2 units. We construct a feasible solution Q_2', F_2', s_2' of 3PTxl and denote its objective value as $\Pi^{k'}$. First, round down Q_2 to the nearest breakpoints k_j value and denote it as Q_2'. Second, let $F_2' = F_2 - (w_{i'}(Q_2) - w_{i'}(Q_2'))$ and $s_2' = s_2$. Denote the prices paid by each consumer as $p^{k'}(y_i)$. Now, we show that $\Pi^{k'}$ is close to Π^c.

In 3PTc, let $\{i_1', \ldots, i_2'\}$ be the consumer types who purchase exactly Q_2 units, and let $\{i_3', \ldots, I\}$ be those who purchase strictly more than Q_2 units. We first show that, consumers $i \in \{i_1', \ldots, i_2'\}$ who purchase exactly Q_2 units in 3PTc now purchase no less than Q_2' units in 3PTxl, because $w_i(Q_2') - F_2' = w_{i_1'}(Q_2) - F_2 + (w_i(Q_2') - w_{i_1'}(Q_2')) \geq 0$. The equality is by definition of F_2'. The inequality is because i_1' purchases Q_2 at price F_2, and because i is weakly larger than i_1'. Therefore, for consumers $i \in \{i_1', \ldots, i_2'\}$, we have

$$p^c(y_i) - p^{k'}(y_i) \leq F_2 - F_2' = w_{i_1'}(Q_2) - w_{i_1'}(Q_2') \leq k\beta. \tag{D2}$$

For consumers $i \in \{i_3', \ldots, I\}$, each of them purchases units y_i that satisfies $w_i'(y_i) = s_2$ in 3PTc, *i.e.*, she purchases all the units for which her marginal WTP is equal to the overage fee. Now, in 3PTxl, each one purchases units k_j that satisfies $\frac{w_i(k_j) - w_i(k_j - 1)}{k} \leq s_2 < \frac{w_i(k_j+1) - w_i(k_j)}{k}$. Since $w_i(\cdot)$ is concave, it follows that $y_i - w_i(k_j) \leq k$. Therefore, for consumers $i \in \{i_3', \ldots, I\}$, we have

$$p^c(y_i) - p^{k'}(y_i) \leq ks_1 \leq kw_I'(0) \leq k\beta. \tag{D3}$$

Now consider the following chain of inequalities,

$$\Pi^c - \Pi^k \leq \Pi^c - \Pi^{k'} = \sum_{i=i'}^{I} \alpha_i (p^c(y_i') - p^{k'}(y_i')) \leq k\beta \sum_{i=i'}^{I} \alpha_i \leq \epsilon.$$

The first inequality is because Π^k represents the optimal profit of 3PTxl while $\Pi^{k'}$ represents one feasible profit of 3PTxl and the second is by (D2) and (D3). The first

equality is by the definitions of $\Pi^{k'}$ and $\Pi^{c'}$, the second is by $\sum_{i=1}^{I} \alpha_i$, which is normalized to one, and the last is by the definition of ϵ. □

Appendix E: Proof of Proposition 5

Proof Let a 3PTg problem be problem A and its optimal value be Π^g. Now, for a given interval $(\theta^l, \theta^h]$ where $\underline{\theta} \leq \theta^l \leq \theta^h \leq \bar{\theta}$, we replace $w(\theta, y)$ of all consumer types $\theta \in (\theta^l, \theta^h]$ with $w(\theta^l, y)$ in problem A to get a Problem B. Let F', Q' and s' be the optimal solution of problem B, and Π^{gl} be its optimal value. Now we adopt F', Q' s' in problem A and denote its objective value as $\Pi^{g'}$. Obviously, since F', Q' and s' do not change, $p(y)$ remains the same in problem A as that in problem B. Since the WTPs for consumers in $\theta \in [\underline{\theta}, \theta^l] \cap (\theta^h, \bar{\theta}]$ do not change either, they will make the same purchasing decision in both problems and thus, seller's profit for these consumers does not change. For any consumer $\theta \in (\theta^l, \theta^h]$, she prefers $y^*(\theta^l)$ to any $y^*(\theta_1)$ for $\theta_1 \in [\underline{\theta}, \theta^l]$ because

$$\begin{aligned} w(\theta, y^*(\theta_1)) - p^*(\theta_1) \leq w(\theta^l, y^*(\theta^l) - p^*(\theta^l) + (w(\theta, y^*(\theta_1)) \\ -w(\theta^l, y^*(\theta_1))) \leq w(\theta, y^*(\theta^l)) - p^*(\theta^l), \end{aligned} \tag{D4}$$

where the first inequality holds because consumer θ^l prefers $y^*(\theta^l)$ to $y^*(\theta_1)$, and the second inequality holds because $w(\theta^l, y^*(\theta^l)) - w(\theta^l, y^*(\theta_1)) \leq w(\theta, y^*(\theta_1)) - w(\theta, y^*(\theta^l))$ by SCP. It means that consumer θ always buys more than $y^*(\theta^l)$. Now consider the following chain of inequality:

$$\begin{aligned} \Pi^{gl} &= \int_{\underline{\theta}}^{\theta^l} f(\theta) p^*(y^*(\theta)) d\theta + \int_{\theta^l}^{\theta^h} f(\theta) p^*(y^*(\theta^l)) d\theta + \int_{\theta^h}^{\bar{\theta}} f(\theta) p^*(y^*(\theta)) d\theta \\ &\leq \int_{\underline{\theta}}^{\theta^l} f(\theta) p^*(y^*(\theta)) d\theta + \int_{\theta^l}^{\theta^h} f(\theta) p^*(y'(\theta^l)) d\theta + \int_{\theta^h}^{\bar{\theta}} f(\theta) p^*(y^*(\theta)) d\theta \\ &= \Pi^{g'} \leq \Pi^g. \end{aligned}$$

The first equality holds because within this optimal solution $y^*(\theta)$, we have $y^*(\theta) = y^*(\theta^l)$ for $\theta \in (\theta^l, \theta^h]$, *i.e.*, consumers within the given interval all purchase the same choice as consumer θ^l does, because their WTP functions are all identical. The first inequality holds because of Equation (D4). The second equality holds by definition of $\Pi^{g'}$. The last inequality holds because $\Pi^{g'}$ is the objective value of a feasible solution for problem A while Π^g is its optimal value. Thus, we have proved $\Pi^{gl} \leq \Pi^g$ for 3PTg. By using the same type of arguments, it is straightforward to prove that, when replacing $w(\theta, y)$ of all consumer types $\theta \in (\theta^l, \theta^h]$ with $w(\theta^l, y)$, we achieve an overestimation of the optimal value of 3PTg. □

Appendix F: Proof of Theorem 3

Proof We first prove $\Pi^{gd} - \epsilon \leq \Pi^g$ in two steps. In step one, we show that $\Pi^{gd} - k\beta \leq \Pi^m$, where Π^m is the optimal value of a 3PTc problem in which consumers' WTPs are $w^m(\theta, y) = w(m_i, y)$ for $m_i \leq \theta < m_{i+1}$. Then, we get $\Pi^{gd} - k\beta \leq \Pi^m$ directly following Theorem 2. In step two, we show that $\Pi^m \leq \Pi^g$. Construct a sequence of 3PTg problems $Q^i, i \in \{0, 1, \ldots, I\}$. In Q^0, consumers' WTPs are $w(\theta, y)$ from the original 3PTg problem and thus, the optimal value of Q^0 is Π^g. In $Q^i, i \in \{1, \ldots, I\}$, consumers' WTPs are the same as those in Q^{i-1} for $\theta \in [\underline{\theta}, m_i] \cup (m_{i+1}, \bar{\theta}]$, and are placed with $w(m_i, y)$ for $\theta \in [m_i, m_{i+1})$. Clearly, the optimal value of Q^I is Π^m. Let Π^{Q_i} be the optimal value of Q_i. Following Proposition 5, we have $\Pi^g = \Pi^{Q_0} \geq \Pi^{Q_1} \geq \ldots \geq \Pi^{Q_I} = \Pi^m$. Thus, it is proved that $\Pi^{gd} - k\beta \leq \Pi^g$.
Next, we prove that $\Pi^g \leq \Pi^{gd} + \epsilon$ in two steps. In step one, we prove that $\Pi^g \leq \Pi^m + m\beta$. We still define problems Q^i as above. In addition, we define problems $P^i, i \in \{0, 1, \ldots, I\}$ as: In P^0, consumers' WTPs are $w(\theta, y)$ from the original 3PTg problem and thus, the optimal value of P^0 is Π^g. In $P^i, i \in \{1, \ldots, I\}$, consumers' WTPs are same as those in P^{i-1} for $\theta \in [\underline{\theta}, m_i] \cup (m_{i+1}, \bar{\theta}]$, and are placed with $w(m_{i+1}, y)$ for $\theta \in [m_i, m_{i+1})$. Observe that

$$\Pi^{P_i} - \Pi^{Q_i} \leq \Pi^{P_{i-1}} - \Pi^{Q_{i-1}} + (F(m_{i+1}) - F(m_i))m\beta.$$

The inequality holds because the incremental revenue by increasing WTP for each consumer $\theta \in [m_i, m_{i+1})$ from $w(m_i, y)$ to $w(m_{i+1}, y)$ cannot be more than the change in WTP, $w(m_{i+1}, y) - w(m_i, y)$, which is no more than $m\beta$. Therefore, we have the following result:

$$\Pi^g \leq \Pi^m + \sum_{i=1}^{I} ((F(m_{i+1}) - F(m_i))\, m\beta = \Pi^{gd} + k\beta + m\beta.$$

□

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